

TABLE OF CONTENTS

S-001 - GENERAL DESIGN REQUIREMENT OF STRUCTURE, CIVIL AND BUILDING WORKS

	Page No.
1. GENERAL	S 001-3
2. CODES AND STANDARDS	S 001-3
3. SCOPE OF STRUCTURE, CIVIL AND BUILDING WORKS	S 001-3
3.1 Soil Investigation.....	S 001-3
3.2 Field and Laboratory Testing.....	S 001-4
3.3 Surveying and Setting Out	S 001-4
3.4 Site Preparation	S 001-5
3.5 Piling Works.....	S 001-5
3.6 Foundation Works	S 001-6
3.7 Earth Works	S 001-6
3.8 Dredging (Not Applicable)	S 001-6
3.9 Structural, Architectural and Building Facility Work	S 001-6
3.10 Other Civil Work.....	S 001-7
4. CONCEPTUAL DESIGN OF STRUCTURE, CIVIL AND BUILDING WORKS	S 001-9
4.1 The Gas Turbine Building (Not Applicable)	S 001-9
4.2 Gas Turbine Foundation (Not Applicable)	S 001-9
4.3 Steam Turbine Building.....	S 001-9
4.4 Steam Turbine and Generator Pedestal.....	S 001-10
4.5 Heat Recovery Steam Generator (HRSG)	S 001-11
4.6 Stack for HRSG.....	S 001-12
4.7 Central Control Building (for Block 3 and Block 4).....	S 001-12
4.8 Miscellaneous Buildings and Structures.....	S 001-13
4.9 Foundation and Structure for BOP.....	S 001-14
4.10 Electrical Structures.....	S 001-15
4.11 Circulating Cooling Water System.....	S 001-15
4.12 Other Structures.....	S 001-17
5. DESIGN REQUIREMENT	S 001-17

5.1	Structural Design Requirements.....	S 001-17
5.2	Architectural Design Requirements	S 001-21
5.4	Other Civil Design Requirements	S 001-23
6.	QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)	S 001-25
7.	METHOD OF MEASUREMENT OF PAYMENT	S 001-25

S 001- GENERAL DESIGN REQUIREMENT OF STRUCTURE, CIVIL AND BUILDING WORKS

1. GENERAL

- 1.1 The site layout drawing in the Bid Drawings shows the Owner's intended layout of the Muara Tawar Gas Fired Power Plant Block 2, 3 and 4 Add-on Project (the Project).
- 1.2 The above layout has been prepared on the basis of three (3) combined cycle blocks. Block-2 Add-on is consisting of two (2) HRSG and one (1) steam turbine generator. Each Block 3 and 4 Add-on is consisting of three (3) HRSG and one (1) steam turbine generator.
- 1.3 The Bidder is required to strictly comply with the location and arrangement of the major buildings, structures and other facilities as shown on the Bid Drawings.
- 1.4 The location and arrangement of the major buildings, structures and other facilities as shown on Bid Drawings shall not be altered unless approved by the Owner.
- 1.5 Unless otherwise noted, the architectural views of the buildings shall be designed in accordance with the Bid Drawings.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of "Codes and Standards" stipulated in hereinafter.
- 2.2 The Codes and standards of the various organizations extracted from the PART 4, Sub-section 2.5 "Codes and Standards" are specifically applicable to the design, manufacture and testing of the works and are included in the respective Technical Specifications.

3. SCOPE OF STRUCTURE, CIVIL AND BUILDING WORKS

3.1 Soil Investigation

- 3.1.1 The Bidder shall perform soil investigation to confirm and verify the existing actual soil conditions for design and construction works. The cost for soil investigation shall be included in the Bid Prices.
- 3.1.2 The geo-technical data for the existing plant (Muara Tawar Gas Turbines Power Plant, Block 1 and 2) and Extension Project (Block 5), those which have been prepared by the Owner, are compiled in the Book IV Bid Drawings for reference.

- 3.1.3 The Bidder shall submit a soil investigation program to the Owner for approval immediately after Contract Award.
- 3.1.4 The soil investigation shall consist of mobilization, demobilization, daily cleaning of work area, transports and furnishing of labors, all machines, equipment and installations necessary to perform the soil investigations.
- 3.1.5 The Bidder shall submit the soil investigation report which includes a recommendation of foundation type for all buildings and structures required in the Project as a draft form. The final report shall be submitted after draft report has been approved by the Owner.
- 3.1.6 The approved draft report shall not relieve the Bidder from his obligation to undertake additional soil investigations, if the Owner requires another soil investigation under consideration of the result of the previous soil investigation. In this case, the Bidder shall undertake additional soil investigation at the account of the Bidder.

3.2 Field and Laboratory Testing

- 3.2.1 The Bidder shall appoint at his account an approved independent company to carry out all testing required in this Technical Specifications for the work under the scope of this Contract as indicated in Part 4, Subsection 1.2 “Summary of Work”.
- 3.2.2 The Bidder shall furnish labor, qualified technicians, all materials, and testing equipment/apparatus to perform the field and laboratory testing.
- 3.2.3 The field and laboratory testing shall consist of carrying out all tests for piles, earthwork, structural steel work, welding, concrete work and any other tests to perform work under the scope of this Contract and as required in this Technical Specifications herein, including keeping the records and submitting it to the Owner.

3.3 Surveying and Setting Out

- 3.3.1 The Bidder shall conduct surveying work on the basis of the existing bench mark which has been provided by the Owner.
- 3.3.2 The Bidder shall perform all surveying and setting out and calculation necessary to establish the accurate location of the Plant in accordance with the General Arrangement Drawings as stated in Subsection 2.2 “Data and Drawing Submission”.

3.3.3 The surveying work shall include furnishing all labor, technicians, qualified surveyor, and required surveying equipment and apparatus.

3.3.4 Calibration certificate shall be provided for the survey equipment used.

3.4 Site Preparation

3.4.1 The site preparation work shall comprise of the following works:

- 1) Provision of temporary facilities, consisting of temporary site office for the Owner and for the Bidder, temporary roads and parking, temporary water supply system, temporary electrical supply system, temporary telecommunication system, drainage ditches, temporary fence between the existing Plant area and the Project area, preparation for temporary laydown area, survey of existing underground structure and utilities, and other temporary work required by the Bidder or as directed by the Owner.

The location of the temporary facilities mentioned above will be allocated by the Owner at the commencement of the construction.

- 2) Land reclamation including stripping, grading and levelling for the final ground level of the Plant (GL), EL + 2.80 m as stipulated in Sub-section 1.1 of Part 4.
- 3) Demolition of the structures, foundations, roads, fence, drainage ditches, etc. which are located in the Project area and required to be removed. Disposal of the demolished structures, debris and materials shall be the responsibility of the Bidder and at the account of the Bidder.

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3.5 Piling Works

3.5.1 The Bidder shall design and provide a circular hollow prestressed concrete pile foundation to support buildings, structures and equipment, which are required as shown on the Bid Drawings and as described in the Sub-section 1.2 Summary of Work. The piling construction activities shall not disturb the operation of existing power plant.

3.5.2 The piling works shall include mobilization, demobilization, transport, and furnishing material, labors, all machines and equipment and installations necessary to execute the piling works including loading test and its analysis in accordance with the Technical Specification S-320 hereinafter and the piling record and the result of the loading test shall be reported to the Owner for approval.

3.5.3 The piling works shall also include settlement and vibration monitoring during construction.

3.6 Foundation Works

- 3.6.1 The Bidder shall design and provide complete reinforced concrete foundations for buildings, structures and equipment, which are required as shown on the Bid Drawings and as described in the Sub-section 1.2 Summary of Work.
- 3.6.2 The foundation works shall include, but not limited to, the following works:
- 1) Excavation, dewatering, foundation preparation, backfill and compaction
 - 2) Sand bedding, lean concrete and mat concrete
 - 3) Formwork and reinforcing steel works
 - 4) Miscellaneous embedded steel
 - 5) Anchor bolts
 - 6) Grouting
- 3.6.3 The foundation works shall also include grounding and any other works necessary for the required buildings, structures and equipment.

3.7 Earth Works

- 3.7.1 The earth works shall consist of cleaning, stripping, grubbing, excavation and filling, which include compacting, leveling, protection, sheeting, shoring, bracing and cofferdams, dewatering and disposal of excavated material to the designated place where is approximately 10 km far from the site.
- 3.7.2 The Bidder shall perform all the earth works necessary in constructing foundations and other civil work, including furnishing labor, equipment, technicians and a qualified professional supervisor.
- 3.7.3 The Bidder shall be responsible for maintenance of the designed ground level as shown on the Bid Drawings, and this work shall include the site filling work in case that settlement occurs at the site due to the site preparation work during construction period.

3.8 Dredging (Not Applicable)

3.9 Structural, Architectural and Building Facility Work

- 3.9.1 The Bidder shall design and construct complete structure, architectural and building facility work, including equipment supports, piping supports and etc. as required in Part 4, Subsection 1.2 “Summary of Work” and as shown on the Bid Drawings.
- 3.9.2 Structural work shall include, but not limited to, concrete structure and the steel structure with bolting and welding connection and complete gallery works including column, base plates, floor and roof beams, plate girders, vertical and horizontal bracing, floor slabs, walls, stairs, checkered plates, grating, hand railing, floor loading signs, door frames and all necessary

connections, floor metal deck, miscellaneous steel and studs embedded parts, anchor bolts, grouting, shop prime painting, field finish painting and any other necessary items to perform the required structural works. The structural requirement shall be referred to APPENDIX-2 “Building Structure List”.

3.9.3 The architectural work shall include, but not limited to, roof, all exterior and interior wall siding, masonry walls, doors and windows, ceilings, louvers, gutters and downspout, lockers, kitchen appliances, etc. The finishing schedule for exterior and interior building shall be referred to APPENDIX-1 “Building Finishing Schedule”

3.9.4 The building facility work shall include, but not limited to, water supply system, drainage system, sanitary fixture, power supply system, communication system, lightning protection system with grounding, unless otherwise indicated in APPENDIX-3 “Building Facility List” and APPENDIX-4 “Service Water Supply & Drainage Concept”

The following building facility works shall be referred to the respective Technical Specifications:

- 1) Power supply system: E-001
- 2) Communication system: E-271 (consisting of telephone system and paging system, etc.)
- 3) Lightning protection system with grounding: E-241

3.9.5 The works for structure, architecture and building utility shall include mobilization, demobilization and furnishing all materials, labors, equipment and installation necessary to perform the complete structure, architecture and building utility works.

3.10 Other Civil Work

3.10.1 Permanent On-Site Road and Parking Area

- 1) The work shall include demolish and renovation of the existing road and construction of new on-site road in the Add-on area as shown on the Bid Drawings.
- 2) Any additional road required for operation and maintenance access of equipment supplied by the Bidder and as result of approved detail design performed by the Bidder which is not foreseen during preparation of this Bid Proposal shall be included in the scope of the Bidder’s work without any additional cost to the Owner.

3.10.2 Drainage Works

- 1) The Bidder shall design and construct complete drainage system as required by the Plant such as storm water drainage, oily water drainage, waste water drainage, sanitary water drainage.

- 2) The storm water drainage system shall be provided in accordance with the Technical Specification S-600 as shown on the Bid Drawing, indicating only the main storm water drainage system with consideration of connecting to the existing drainage system. Other storm water drainage line from every area shall be developed and routed to the above drainage system.
- 3) Sanitary water drainage system shall be provided to the building which has toilet, kitchen or shower. It shall be terminated in nearby septic tank before discharge into the storm water drainage system.
- 4) Oily water drainage system shall be provided at around transformer area and switchgear area with oil separator before discharge into the storm water drainage system.

3.10.3 Pavement and Fence and Gate Works

The Bidder shall design and construct the following paving work, but not limited to, as follows:

- 1) Gravel Paving

Gravel paving shall be provided in the area of switchyard, waste water treatment building, desalination building, and piping route corridor.

- 2) Concrete Paving

Concrete paving shall be provided in the area of HRSG, by-pass stack, steam turbine building.

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- 3) Fence and Gate

Fencing and gate shall be provided by the Bidder at the perimeter of the following areas:

- Hazardous areas such as chlorination, transformer and gas stations.
- Property line near the intake modification area.

This work shall include design, furnishing all materials, labour and equipment to perform the work.

- 4) Environmental Protection Work

The Bidder shall design and provide the environmental protection work as specified in the Technical Specification S-650 as follows:

- Grass shall be planted in the areas which are not covered by paving work.
- Trees and gardening shall be provided, and a sufficient number of garden house valve boxes shall be furnished by the Bidder to facilitate watering for trees and grass.

4. CONCEPTUAL DESIGN OF STRUCTURE, CIVIL AND BUILDING WORKS

For carrying out the further design of the structure, civil and building works required in the Project, the Bidder shall keep the following concept in mind.

The architectural feature in general for the Power Plant shall be ensured in good appearance as well as adequate protection against sun, wind, rain and sea water injury.

4.1 The Gas Turbine Building (Not Applicable)

4.2 Gas Turbine Foundation (Not Applicable)

4.3 Steam Turbine Building

4.3.1 The Steam Turbine Building will be one (1) building for The Block 2, 3 and 4. The building shall have minimum size of 170 m in the longitudinal direction and 35 m in the transversal direction as shown on the Bid Drawings to house three (3) steam turbine generator and all associated mechanical and electrical equipment, and shall have ample space for maintenance specified in the respective Technical Specification for the Mechanical and Electrical Works. The existing steam turbine building will be modifying to allow access way during operation and maintenance.

4.3.2 The building shall be designed as steel structure supporting overhead crane (specified by the Mechanical Specification) for maintenance and enclosed with insulated metal siding and roofing. The existing railway of overhead crane shall be connected to new railway of overhead crane.

4.3.3 Reinforced concrete roofing shall be provided on the floor where the deaerator equipment is installed, if it is recommendable from operational and maintenance point of view.

4.3.4 All bays shall be structurally connected together to obtain the benefit of stiffening against horizontal loading in both the longitudinal and transversal directions. Sufficient bracing shall be provided horizontally at the roof and operating levels, and vertically at selected column bays on the longitudinal direction of the building.

4.3.5 The building shall have a reinforced concrete foundation supported by piles.

4.3.6 The operating floor shall be made of a reinforced concrete, finished with monolithic steel trowel finish with hardener. Steel grating shall be provided for any floor openings for safety purpose.

4.3.7 All electrical panels shall be installed on raised reinforced concrete pads.

4.3.8 Ground floor level of the building shall be designed a minimum 30 cm above the surrounding ground level.

- 4.3.9 Ramp way shall be provided at every main access road to the building.
- 4.3.10 1.00 m high plastered reinforced concrete block wall shall be provided along the periphery of the building.
- 4.3.11 To maximize the benefit of natural light colored aluminum glazed windows shall be provided in the insulated metal siding.
- 4.3.12 Roof extraction vents shall be provided at the roof.
- 4.3.13 Floor drainage system shall be provided with sump pit and pumps.
- 4.3.14 The scope of works for Steam Turbine Building is summarized hereunder:
- Piling
 - Concrete foundation with earth works
 - Structural works
 - Metal siding and roofing
 - Architectural works
 - Gallery works
 - Building facility works
 - Drainage works
 - Modify Existing Building Wall and Roof
 - Other related works
- 4.4 Steam Turbine and Generator Pedestal**
- 4.4.1 Cantilevers should be avoided, but if cantilever elements are required by the machine outline, they should be kept as short as possible and with the ratio of cantilever length to depth not to exceed unity.
- 4.4.2 There shall be complete vibration isolation between the pedestal and entire surrounding turbine building at first floor, mezzanine floor and operating floor. Generator Pedestal shall be separated with Steam Turbine Building Foundation.
- 4.4.3 Permanent elevation reference points (benchmark) shall be embedded in the top deck. These reference points shall be used to monitor the deformations of the pedestal, subsequent to the alignment of the turbine shaft, throughout the service life of the machinery.
- 4.4.4 The benchmark shall be provided for adequate protection to ensure that reading accuracy can be obtained within 0.025 mm.

- 4.4.5 A topping slab shall be provided on top of deck with 75 mm to 150 mm thick. The slab shall be reinforced with welded wire mesh and finished with steel trowel with control joints to minimize cracking.
- 4.4.6 The scope of works for Steam Turbine and Generator Pedestal is summarized hereunder:
- Piling
 - Concrete mat foundation and concrete pedestal with earth works
 - Anchor bolt
 - Grouting
 - Embedded material
 - Other related works
- 4.5 Heat Recovery Steam Generator (HRSG)**
- 4.5.1 The Bidder shall design and provide a complete supporting steel structure with concrete mat foundation supported by piles.
- 4.5.2 Elevated platform with galvanized grating, 40 mm thick shall be provided at HP & LP drum level adjacent to the HRSG structure.
- 4.5.3 Stairs shall be provided for access from ground level to HP & LP drum level and from this level to the lowest platform level provided at every HRSG stack.
- 4.5.4 Platform shall be designed and installed at the top of HRSG stacks for installation and maintenance obstruction lights.
- 4.5.5 HP & LP drum platforms shall be connected together with access gallery to provide access between adjacent HRSG's, within each Block.
- 4.5.6 Un-insulated metal roof shall be provided to cover the LP and HP drum platforms.
- 4.5.7 An elevator will be provided for each HRSG area to service various levels up to the top of HRSG. The elevator shaft will be completely enclosed with brick wall. Elevator landings will be designed as reinforced concrete floor and will be provided at the various stop levels. The elevator machine room will be located at grade floor level adjacent to the elevator shaft.
- 4.5.8 The scope of works for Heat Recovery Steam Generator (HRSG) is summarized hereunder:
- Piling
 - Concrete mat foundation with earth works
 - Other related works

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- Miscellaneous equipment foundation required in the area of HRSG

4.6 Stack for HRSG

- 4.6.1 Stacks shall be provided with concrete foundation supported on piles. Platform and ladder shall be provided for access from the ground level to the top of the stacks to install and maintain obstruction lights.
- 4.6.2 The scope of works for Bypass Stack is summarized hereunder:
- Piling
 - Concrete mat foundation with earth works
 - Other related works
 - Miscellaneous equipment foundation required in the area of Gas Turbine

4.7 Central Control Building (for Block 3 and Block 4)

- 4.7.1 The building shall be located closed to Steam Turbine Building with three (3) stories.
- 4.7.2 The Bidder shall notify the Owner after the Contract has been signed the time when the control room should be available.
- 4.7.3 The ground floor elevation shall be designed 30 cm above ground level.
- 4.7.4 The building shall be made of reinforced concrete structure with concrete foundation supported by piles.
- 4.7.5 The roof of the building shall have a reinforced concrete slab structure with built-up roofing to support equipment such as the domestic water storage tank and the air conditioning cooling system.
- 4.7.6 All architectural works inclusive of floor tile, ceiling and doors shall be designed in accordance with Design Requirement provided hereinafter.
- 4.7.7 The free access floor shall be provided at central control room, and computer room at 50 cm above the structural concrete slab level.
- 4.7.8 An air conditioning system shall be provided in the majority of rooms such as control room, office & meeting room, computer room, and electrical room, and others. Exhaust vents for ventilation shall be provided for kitchen and toilet.
- 4.7.9 The scope of works for Central Control Building is summarized hereunder:
- Piling
 - Concrete foundation with earth works
 - Structural works

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- **Concrete / Concrete Block Wall / Brick Wall**

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- Built –up roofing
- Architectural works
- Gallery works
- Building facility works
- Drainage works
- Other related works
- Miscellaneous equipment foundation in the area of Steam Turbine

4.8 Miscellaneous Buildings and Structures

4.8.1 The following miscellaneous buildings shall be designed and constructed by the Bidder under this Contract.

- 1) Storage room, Workshop and Laboratory Building
- 2) Guard house (Location shall be decided at later stage)
- 3) Emergency diesel generator building with equipment foundation
- 4) Water treatment control building with equipment foundation
- 5) Waste Water treatment control building with equipment foundation
- 6) Desalination control building with equipment foundation
- 7) Storage Building for B3 Waste and Material
- 8) Cooling water pump house and pump pit with equipment foundation
- 9) Firefighting pump house
- 10) Hydrogen building
- 11) Other electrical building related to the work as required

4.8.2 The above buildings shall be designed as reinforced concrete structure or steel structure as indicated in APPENDIX-2 “Building Structure List”, and supported on piles.

4.8.3 All the buildings shall be finished in accordance with “Building Finishing Schedule” attached in APPENDIX-1”.

4.8.4 All the buildings shall be provided with outside ladders for access from ground level to rooftop.

4.8.5 Floor drainage shall be provided in the emergency diesel generator building and shall be routed through the oil separator before discharging to the storm water drainage system.

4.8.6 Shelters shall be provided for the following equipment and plant:

- 1) Water treatment plant
- 2) Waste water treatment plant

4.8.7 All shelters shall be of steel structure with un-insulated metal roofing. Bracing shall be provided for structural stability. The size of shelter shall be adequate to cover all equipment. Floor shall be of graded concrete floor with steel trowel finish.

4.8.8 Workshop and laboratory building shall be designed 2 (two) stories. First floor used for workshop and storage room, second floor for laboratory room. The ground floor for workshop shall be made of reinforced concrete slab on grade and have control joints to avoid from any cracks occurring by shrinkage.

4.8.9 The scope of works for the above Miscellaneous Buildings is summarized hereunder:

- Piling
- Concrete foundation with earth works
- Structural works
- Metal siding
- Gallery works
- Architectural works
- Building facility works
- Drainage works
- Other related works

4.9 Foundation and Structure for BOP

4.9.1 The following foundation and structure for Balance of Plant (BOP) shall be designed and constructed by the Bidder under this Contract.

- 1) Chlorination plant: including storage tank, piling, concrete foundation, fencing surrounding the storage tank and other related works
- 2) Desalination system: including piling, concrete foundation, waste water pit and other related works
- 3) Waste water treatment plant: including piling, concrete foundation, unit neutralizing pit, waste water storage pond, pH control oxidation pit and mixing pit, coagulation- sedimentation pit, clear water pit, neutralizing pit, purified water waste pit, sludge enrichment pit, sludge storage pond, waste water pit, and other related works.
- 4) Water Treatment plant: Piling, Water Treatment Control Building, Water tank foundation, Acid storage tank foundation, Concrete foundation, ex

cavation and back fill, Structural Works, Gallery Works, Architectural Works, Building Facility Works, Other related works.

- 5) Raw water tank foundation: including piling, concrete foundation, and other related works
- 6) Make-up water Tank foundation: including piling, concrete foundation, and other related works

4.10 Electrical Structures

4.10.1 The following electrical structures shall be designed and constructed by the Bidder under this Contract.

- 1) Transformer and Auxiliary Transformer: including piling, concrete foundation, fire wall, drainage, oil pit, gravel finishing, fencing and other related works. Oil separator shall be provided nearby the transformer foundation.
- 2) Isolated phase bus duct structures for Steam Turbine Building to Switchyard: including piling, concrete foundation, structural works and other related works
- 3) Switchyard: including piling (if required), concrete foundation, structural works, drainage, fence and gate, maintenance road and other related works
- 4) Underground cable ducts and cable trenches: including piling (if required), concrete foundation, structural works, trench cover, sump pit and other related works
- 5) Miscellaneous equipment foundation required in the area of electrical structure.

4.11 Circulating Cooling Water System

4.11.1 Intake Open Channel

4.11.1.1 Existing Site Conditions and Location of New Facilities

The existing cooling water intake facilities are located north of the Muara Tawar Plant and at the end of the intake channel having 2.8 km long. These facilities are now supplying cooling water to Block 1 only and to Block 5. After review of the existing site conditions new cooling water intake facilities will be tapping at existing intake canal.

4.11.2 Circulating Cooling Water Pump Pit

As specified in the Technical Specification for the Mechanical Works, the C.W. pump shall be indoor type and be installed at the C.W.P. pit. The size of C.W. pit & house shall be decided by the Bidder under consideration of ample space for installation and maintenance.

The pit shall be made of a reinforced concrete structure supported by piles and the CWP house shall be of a steel structure enclosed with un-insulated metal siding and roofing.

The floor level of the operating floor for the C.W pit shall be designed at 30 cm above ground level.

Crane rails shall be fixed to the steel structure for safety operation of overhead crane. The overhead crane shall have suitable lifting capacity for maintenance purpose. The existing railway of overhead crane shall be connected to new railway of overhead crane.

Hot dipped galvanized steel handrail shall be provided at the perimeter of the C.W. pit adjacent to the intake channel.

Stop logs with storage racks and lifting beam shall be provided.

Any miscellaneous steel and ladders installed below water level shall be stainless steel material.

4.11.3 **Earthworks for Cooling Water Supply Pipe Line**

The cooling water supply pipe shall be installed and connected to condenser in the steam turbine building. The scope of the Civil Works shall cover the earthworks only. The supply and installation of the cooling water pipe shall not be included in the Civil Works.

For carrying out the earthwork, the Bidder shall pay particular attention for the crossing area of the existing discharge tunnel for Block 1 and this cooling water pipe line.

4.11.4 **Discharge Tunnel**

A reinforced concrete tunnel shall be designed and constructed by the Bidder to discharge the cooling water from the condenser to the open discharge channel.

The tunnel shall be supported by piles.

Expansion joints shall be provided with spacing of joints.

Manholes including stainless steel ladders shall be provided at every 50 m (maximum) for access to inside of the tunnel.

Any miscellaneous steel or embedded steel installed below water level shall be of stainless steel material.

Weir and drop structure including stop logs shall be provided at the end of tunnel.

4.11.5 **Discharge Open Channel**

New discharge tunnel shall be connected to the existing open channel by modifying the existing structure. The required modification for wall height of the existing discharge open channels shall be included in the scope. The construction as connecting the new discharge tunnel to the existing open channel shall not disturb the operation of the existing open channel.

4.12 Other Structures

4.12.1 Equipment Foundation

Foundation for any mechanical and electrical equipment shall be designed as reinforced concrete foundation supported by piles.

No expansion anchors shall be applied for vibratory equipment.

The top of all foundations shall be kept 30 cm above the ground level and have a proper slope to prevent any standing water.

4.12.2 Cable Trenches and Cable Ducts

Cable trenches and cable ducts shall be of reinforced concrete placed in situ, including pipe sleeves, if required. All trenches shall be covered with removable reinforced concrete covers.

All trenches shall be provided with drainage system, including sump pits at every 50 m distance.

4.12.3 Take Off Structures in Switchyard

All take off structures, gantry towers, etc. for supporting electrical equipment shall be galvanized steel and supported by piles.

All areas of switchyard shall have gravel paving to minimize the potential of grass growth.

4.12.4 Pipe Rack Structure

Structure for pipe rack shall be designed as steel structure and the foundation thereof shall be designed as reinforced concrete foundation supported by piles.

The top of all foundations shall be kept 30 cm above the ground level and have a proper slope to prevent any standing water.

5. DESIGN REQUIREMENT

5.1 Structural Design Requirements

- 1) The design of piles and foundations shall be based on the data obtained from soil investigation which will be carried out by the Bidder. The geotechnical data for the existing plant (Muara Tawar Gas Turbines Power

Plant, Block 1 and 2) and Extension Project (Block 5) which are compiled in the Book IV Bid Drawings can be used as the reference.

- 2) Unless otherwise required by the manufacturer's equipment specifications, the total settlement of structures shall not exceed 25 mm.
- 3) The supporting structural steel for all buildings and structures constructed by the Bidder shall be of adequate strength, stiffness and rigidity, and shall be adequately braced for the loads to which the steel will be subjected, and shall be completely suitable for the service.
- 4) All structures will be designed by ACI for concrete structure and AISC for Steel Structure, or AIJ (Architectural Institute of Japan) for concrete and steel structure, or applicable Codes of Practice in Indonesia.
- 5) Steel work design shall be designed by the working stress method and to be carried out in accordance with AISC or the appropriate Indonesian standards. Reinforced concrete design shall be designed by the ultimate strength method and to be carried out in accordance with ACI318 or appropriate Indonesian standards and codes of practice approved by the Engineer.
- 6) The structural design shall conform to the applicable requirement of the codes and standards referenced above.
- 7) All Bidder's design and detailing shall be approved by the Owner prior to commencement of fabrication and construction.

5.1.1 **Loading Criteria**

Structural elements shall be designed by applying the following loading criteria:

- 1) Dead Load (DL)
- 2) Live Load (LL)
- 3) Equipment Load (EL)
- 4) Piping Loads (PL)
- 5) Cable Loads (CL)
- 6) Installation Loads (IL)
- 7) Lay Down Loads (LD)
- 8) Crane/ Hoist Load (C.O.L, C.D.L)
- 9) Wind Load (WL)
- 10) Seismic Load (SL)
- 11) Soil Pressure(SP)
- 12) Construction Load (CN)

The details for the above loading criteria shall be referred to S-002 Loading Criteria.

5.1.2 Particular Attention on Structural Design

5.1.2.1 Pile and Foundation

- 1) Piles shall be supported on bearing layer and shall be considered as end bearing and the friction developing along the shaft in the soft soils shall be neglected.
- 2) According to the Owner's preliminary soil investigation as shown on the Bid Drawings, the bearing layer with more than SPT 50 is situated at about 26 m below the original ground level. But, this is resulted from the only three places of soil investigation work at the limited area.
- 3) Therefore, the Bidder shall convince himself by executing further soil investigation and provide interpretation of the obtained data for confirmation of the pile bearing layer at his own cost.
- 4) Consideration of settlement is expected to occur due to embankment work conducted at the Site and negative friction shall be considered by the Bidder in the design of pile.
- 5) The effect due to the seasonal ground water level shall be taken into account on foundation design.
- 6) The piles and foundation shall have minimum safety factor of 2.5 against bearing capacity failures and the foundation shall not undergo excessive displacement under working loads or exceed specified limits set by the structural tolerance of the supported structure or set by equipment manufacturer, in case of equipment foundation.

5.1.2.2 Superstructure

- 1) All structures shall be designed to support the above imposed loads.

Design, construction and erection of steel structure shall conform to the requirement of AISC, ACI, AIJ or equivalent.

- 2) The design requirement for structures is detailed in the following:

The depth of fully stressed beams and girders shall not be less than $F_y/56000$ (F_y in kg/cm^2) times span unless allowable stresses are reduced. The corresponding maximum L/D ratio for A36 shall be 22 for simple supported members. The L/D ratio shall not exceed 30 under any circumstances.

- 3) For cantilevered beams, the maximum L/D ratio shall not exceed 5.

- 4) For strut members, i.e. members subject to axial load and bending due to self-weight only, the L/D ratio shall not exceed 22.
- 5) For flexural members supporting reciprocating or rotating machinery, the maximum L/D ratio shall be 12.
- 6) The deflection of beams, girders under normal load shall not exceed 40 mm or L/300 whichever is less.
- 7) All connections shall be shop welded and field bolted and shall conform as a minimum to AISC Manual Tables II, III and IV unless otherwise noted.
- 8) The design of steel connections shall be based on Friction Type connections with ASTM A325 or A490 bolts.
- 9) The maximum slenderness ratio (KL/r) of the steel columns shall be limited to 200.
- 10) The steel lateral load-resisting system shall be analyzed and designed to function independently from shear-resisting concrete or concrete masonry exterior and interior walls, and concrete floor slabs acting as shear-resisting diaphragms.
- 11) The lateral deflection or drift of a story relative to its adjacent stories due to seismic event shall not exceed 0.005 H_{sx} as defined in the U.B.C. H_{sx} is the story height of the vertical bracing panel.
- 12) The maximum drift due to wind load shall be limited to 0.002H, where H is the building height below the level under consideration.
- 13) The bracing system should be laid out with regard to permanent and temporary access requirements for equipment removal purpose.
- 14) The size of gusset plates and wing plates for bracing connections shall be reviewed to avoid interferences with piping, cable trays, ducts, etc.
- 15) The horizontal deflection of girder due to wind load shall be limited to 1/180 of the span. For girder supporting windows, the allowable horizontal and vertical deflections shall be consistent with the manufacturer's requirement (typically 6 mm for glass windows). The vertical deflection of the carrier beam shall be limited to the lesser of 25 mm or 1/300 of the span.
- 16) Grating, checkered steel plate or roof metal deck shall not be considered as lateral support for the floor beam.
- 17) The minimum access gallery shall be 75 cm wide unless noted otherwise. A minimum headroom clearance of 2.1m shall be maintained over all galleries. All grating shall be galvanized and suitable for outdoor environment.

- 18) The column base plates shall be designed so that the bearing stresses on concrete foundations do not exceed the allowable value for foundation concrete based on 28 days strength.
- 19) All column base plates shall be founded on concrete foundations at grade level unless noted otherwise.
- 20) The metal deck roof pitches shall be minimum 5 degree inclination, subject to the roofing materials. All roof water shall be collected at edge gutter and drained downward through downspout and discharged to the drainage ditch.
- 21) All concrete work shall be designed in accordance with ACI 318.

5.2 Architectural Design Requirements

5.2.1 Metal Siding and Metal Roof Deck

- 1) The metal siding and metal roof deck for the building shall conform to the Technical Specification provided hereinafter.
- 2) Insulated metal siding and insulated roof deck shall be provided for the Steam Turbine Buildings and shall be designed to obtain thermal and noise reduction in accordance with the related Technical Specification.
- 3) The colour of metal siding and roof deck shall be as specified in the Technical Specification hereinafter or as selected by the Owner.
- 4) All gutters shall be galvanized steel, similarly coated and coloured to match the siding.
- 5) All metal siding and roof shall be designed weather, water and dust tight.

ADD1
-IIG
-006

5.2.2 Doors

- 1) The Bidder shall design and furnish doors as required in the Technical Specification in the Bid Documents. Unless otherwise specified in the Technical Specification in the Bid Documents, all doors shall be steel fabricated at the approved manufacturer. The Bidder shall provide the fire doors with 1.5 hrs fire rated as per NFPA 80. The location for the fire doors shall be conformed to the local regulation “KETENTUAN TEKNIK PENGAMANAN TERHADAP BAHAYA KEBAKARAN PADA BANGUNAN GEDUNG DAN LINGKUNGAN”, March 1, 2000.
- 2) Single doors shall be minimum 0.8 m wide and 2.1m high. Where double doors are required, they shall be sized with consideration of equipment move-in and out.

- 3) Doors more than 2 m wide shall be electric operated roll up doors.
- 4) Vision panels of the doors shall be of clear polished wire glass.
- 5) The doors located in the control room shall be sound proof with a minimum sound transmission class rating of 40 decibels.

5.2.3 Windows

- 1) Window shall be anodized aluminium with clear polished glass except in the Turbine Building where wired glass is provided.
- 2) Windows for the control room shall be insulated glass or double glazing in order to obtain noise and thermal insulation.

5.2.4 Louvers

- 1) Louvers shall be provided at the equipment rooms and other area for ventilation purpose.
- 2) Louvers shall be of anodized aluminium louvers and at exterior location shall be provided with bird screen and air filter.

5.2.5 Floor Finish

- 1) Bidder to provide topping slab over the structural concrete slab shall be provided where switchgear is to be installed or where specifically required for setting of equipment as required by the equipment manufacturer.
- 2) The monolithic steel trowel finish with hardener shall be applied to the operating floor of Steam Turbine Building and other area as specified in the Technical Specification hereinafter.
- 3) Ceramic tile floor shall be applied to the office room, meeting room, kitchen, stairs and other floor as specified in the Technical Specification hereinafter.
- 4) Raised floor shall be provided in the central control room and electronic equipment room.
- 5) Glazed tile shall be applied to the toilet, locker and shower room.
- 6) Multilayer flooring shall be applied in the battery room floor area. The epoxy shall have chemical resistance to alkalis acids solvents, and beverages with material density and bonding strength of not less than 1.3 kg/cm³ and 25 kg/cm² respectively.

5.2.6 Ceiling

- 1) Aluminium louvered ceiling shall be provided in the control room in order to obtain uniform light.

- 2) Suspended acoustic ceiling tile shall be provided in the office area and dining room.
- 3) Plastered board ceiling shall be applied to the kitchen, toilet, locker room and shower room.

5.3 Gallery Works Design Requirements

5.3.1 Ladders

Ladders shall be provided in all pits, manhole and roofs over 6 m above the ground level where there is no stair access.

5.3.2 Handrails

Handrails shall be provided for the following area to protect personnel from drop.

- Around every stairway floor opening and stairway
- Around every floor opening
- Walkway and platform
- Adjacent to dangerous equipment

5.3.3 Other Gallery Works

The following gallery works shall be provided where required.

- 1) Grating
- 2) Stairs
- 3) Checkered steel floor plate
- 4) Guard plate

5.4 Other Civil Design Requirements

5.4.1 On-site Road

- 1) Grade shall be held to a maximum of 8 percent with 6 percent preferable. Minimum radius of curvature shall be 15 m.
- 2) Unless otherwise shown on the Bid Drawings, the main road shall be 8.0 m wide, other road shall be 3, 4 and 6 m wide as shown on the Bid Drawings.
- 3) Loading over culvert and pipes shall be considered in accordance AAHSTO HS20-44 except for areas subject to loading by special heavy haul vehicles.

5.4.2 Storm Drainage

- 1) Runoff resulting from rainfall shall be conveyed to drainage ditches by sloping the tributary surface area.
- 2) The Bidder shall design and provide the drainage system as specified in the Technical Specification hereinafter, unless otherwise shown the Bid Drawings.
- 3) For the storm drainage system, the ditches and culverts shall be adequate to discharge surface run-off due to a 10 years return period. The rational formula shall be used for determining rainfall runoff.
- 4) The ditch and culverts shall be of reinforced concrete construction and designed on the basis of a maximum velocity of 2.5 m/sec..
- 5) The ditch shall be provided with the control box, and the culverts shall be provided with adequate number of manholes with heavy frames and covers.
- 6) Minimum diameter of manhole shall be 1000 mm. Manhole shall be installed at all changes in grade, size or alignment, at all intersections, and at intervals not greater than 50 m.

5.4.3 Transformer Area Drainage & Oil Separator

- 1) To contain accidental oil spills, all drainage from the transformer area shall be routed through an oil separator.
- 2) The design of transformer pit and piping from transformer pit to oil separator shall be based on the largest flow expected from the following conditions unless the Bidder requires larger.
- 3) Surface runoff due to a 10 year recurrence interval from the transformer area.
- 4) Deluge system flow required for the largest transformer fire protection system and shall be based on estimated flow rate of 30 litres/s.
- 5) Accidental oil spill from the largest transformer and shall be based on estimated spilled oil volume of 3000 litres unless the Bidder requires larger.
- 6) The size of main line connecting to the oil separator shall be sized to handle additional flow from other areas as required.

5.4.4 Sanitary Water Drainage

- 1) The Bidder shall design and provide the mechanical sewage system including septic tanks which shall be of reinforced concrete construction with manholes and baffles.

- 2) The piping shall be cast iron or vitreous clay with the minimum size of 150 mm diameter.

- 3) Manholes shall be provided with a minimum clear opening of 900 mm.

Clean outs shall be provided at every 30m and at each change in direction of the sewer system.

- 4) Prior to the design of the drainage absorption field, the Bidder shall perform percolation tests to determine the absorption rate. Based on the field percolation test results, the size of the absorption field shall be determined.

5.4.5 Chemical Waste Water Drainage

The chemical waste water and oily water except Transformer area shall be treated in the Waste Water Treatment Plant (please refer to Mechanical Specification, M-404), and then discharged into the storm water drainage ditch.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Please refer to the “Quality Assurance/Quality Control” and as specifically indicated in the Technical Specifications.

7. METHOD OF MEASUREMENT OF PAYMENT

The payment for Structure, Civil and Building Works shall be done in accordance with the requirement stipulated in Bid Documents. The measurement for progress payment of the Structure, Civil and Building Works shall be based on the following units and quantity installed as shown on the approved Bidder’s drawings:

<u>Items</u>	<u>Units</u>
Survey work	Lump sum
Excavation	Cubic meters
Dredging	Cubic meters (Not Applicable)
Fill work	Cubic meters
Piling	Linear meters
Loading test	Number of loading test
Grating	Square meters
Handrail	Linear meters
Steel structure	Tonnage
Metal deck	Square meters
Anchor bolts	Tonnage
Grout work	Cubic meters
Special coating	Cubic meters

Miscellaneous steel	Tonnage
Painting	Square meters
Concrete	Cubic meters
Reinforcement bar	Tonnage
Formwork	Square meters
Metal siding and roofing	Square meters
Concrete expansion anchor	Numbers
Doors, windows and louvers	Numbers
Floor finish and wall tiles	Square meters
Interior and exterior wall	Square meters
Ceiling	Square meters
Plumbing fixtures	Numbers
Water heater	Numbers
Lockers	Numbers
Floor and roof drain	Numbers
Mechanical sewage system	Lump sum
Septic tank	Numbers
Road work	Square meters
Concrete block paving	Square meters
Gravel paving	Square meters
Fencing	Linear meters
Gabion	Cubic meters
Gates	Numbers
Seeding and sodding	Square meters
Trees and shrubs	Numbers

The price quoted shall include design, detailing, supply, fabrication/erection and any other items required in the respective Technical Specifications.

(Last page of S-001)

APPENDIX-1
BUILDING FINISHING SCHEDULE

			Interior Finishes			Exterior Finish			
No.	Name of Building	Name of Room/ Floor	Floor	Wall	Ceiling / Cover	Roof	Wall	Door	Window
1	Steam Turbine Building	Turbine Hall	Monolithic steel trowel finish with hardener	Double sheet insulated metal cladding/ Cement mortar plaster painted	Galvanized profiled and painted	Lower Roof (If necessary)	Lower Roof (If necessary) Insulated metal cladding + Cement mortar plaster painted	Steel flush door (double swing/ single swing/ roll-up shutter) with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board 6 mm clear polished float glass in board. Aluminium louver with bird screen
		Electrical Equipment Area (LV SWGR Room, MV SWGR Room)	Monolithic steel trowel finish with hardener	Cement mortar plaster painted	Galvanized profiled and painted	Double sheet insulated metal roofing			
		Battery Room	Acid resistant epoxy	Acid Resistant ceramic Tile (up to 1.5 m high) Cement mortar plaster painted (over 1.5 m high)	Noncombustible mineral fibre board with aluminium suspension system	Higher Roof Double sheet insulated metal roofing			
		Auxiliary Equipment Area	Monolithic steel trowel finish with hardener	Double sheet insulated metal cladding	Galvanized profiled and painted				
		Cable Trench	Monolithic steel trowel finish with hardener	Monolithic steel trowel finish with hardener	Checkered steel plate galvanized				
		Emergency Stair, Corridor	PVC tile + Monolithic steel trowel finish with hardener	Cement mortar plaster painted	Fair faced concrete painted				

ADD2
-IIG
-003

No.			Interior Finishes			Exterior Finish			
			Floor	Wall	Ceiling / Cover	Roof	Wall	Door	Window
2	Central Control Building	Central Control Room	Raised floor. (Access floor)	Cement mortar plaster painted Partition wall: glass window	Louvered grid ceiling with aluminium suspension system	Built-up roofing waterproofing + Insulation	Cement mortar plaster painted	Steel flush door with semi-gloss paint finish (soundproof type)	Aluminium window with double glass of 6 mm thick blue tinted polished float glass with reflective out board
		Office and Plant Manager Room, Corridor, Meeting Room, Library, Prayer Room	Homogeneous ceramic tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system				
		Toilet, Shower, Kitchen, Pantry, Locker.	Glazed ceramic tile Unglazed ceramic (for toilet and shower)	Glazed ceramic tile	Moisture resistance ceiling with aluminium suspension system				
		Monitor Room	Raised Floor (Access floor)	Cement mortar plaster painted Partition wall: double glass window	Acoustic tile panel with aluminium suspension system				
		Air Conditioning (AC) equipment Room	PVC tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system				

No.			Interior Finishes			Exterior Finish			
			Floor	Wall	Ceiling / Cover	Roof	Wall	Door	Window
		Cable Trench	Monolithic steel trowel finish	Monolithic steel trowel finish	Checkered steel plate galvanized				
3	Cooling Water Pump House	Electrical Room	PVC tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system	Uninsulated metal roofing	Uninsulated metal cladding + Cement mortar plaster painted	Steel flush door (single swing/ roll-up shutter) with semi-gloss paint finish	
		Plant Room	Monolithic steel trowel finish with hardener	Uninsulated metal cladding	Galvanized profiled				
4	Water Treatment Control Building	Control Room	Ceramic tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system	Uninsulated metal roofing	Uninsulated metal cladding + Cement mortar plaster painted	Steel flush door with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board
		Electrical Room	PVC tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system				
		Toilet	Grazed ceramic tile	Glazed ceramic tile	Noncombustible mineral fibre board with aluminium suspension system				
5	Chlorination Building	Control Room	Monolithic steel trowel with hardener	Acid resistant ceramic tile (up to 1.5m high) Cement mortar plaster painted (over 1.5m high)	Fair faced concrete painted	Built-up roofing waterproofing + Insulation	Cement mortar plaster painted	Steel flush door with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board
		Plant Room (Generator Room)	Acid resistant ceramic tile						
6	Emergency Diesel Generator Building	Plant Room (Generator Room)	Monolithic steel trowel finish with hardener	Cement mortar plaster painted	Fair faced concrete painted	Built-up roofing waterproofing + Insulation	Cement mortar plaster painted	Steel double door with semi-gloss paint finish	Aluminium louver with insulation

No.	Name of Building	Name of Room/ Floor	Interior Finishes			Exterior Finish			
			Floor	Wall	Ceiling / Cover	Roof	Wall	Door	Window
7	Workshop, storage room (first floor) and Laboratory Building (second floor)	Workshop	Monolithic steel trowel finish with hardener	Cement Mortar plaster painted	Galvanized profiled	Built-up roofing water proofing Insulating	Cement mortar plaster painted + Un-insulated metal cladding	Steel flush door (single swing/ roll-up shutter) with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board
		Office	Homogeneous ceramic tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system				
		Toilet	Unglazed ceramic tile	Glazed ceramic tile	Moisture resistance ceiling with aluminium sustension system				
		Laboratory	Acid and Alkaline resistant Floor	Acid resistant ceramic tile (up to 1.5m high) Cement mortar plaster painted (over 1.5m high)	Fair faced concrete painted	Built-up roofing waterproofing + Insulation	Cement mortar plaster painted	Steel flush door with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board
8	Waste Water Treatment Control Building	Control Room	Ceramic tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system	Uninsulated metal roofing	Uninsulated metal cladding + Cement mortar plaster painted	Steel flush door with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board
		Electrical Room	PVC tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system				
		Toilet	Unglazed ceramic tile	Glazed ceramic tile	Noncombustible mineral fibre board with aluminium suspension system				

No.			Interior Finishes			Exterior Finish			
	Name of Building	Name of Room/ Floor	Floor	Wall	Ceiling / Cover	Roof	Wall	Door	Window
9	Desalination Control Building	Control Room	Ceramic tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system	Uninsulated metal roofing	Uninsulated metal cladding + Cement mortar plaster painted	Steel flush door with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board
		Electrical Room	PVC tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system				
		Toilet	Unglazed ceramic tile	Glazed ceramic tile	Noncombustible mineral fibre board with aluminium suspension system				
10	Storage Building for B3 Waste and Material		Acid resistant epoxy	Acid resistant ceramic tile (up to 1.5m high) Cement mortar plaster painted (over 1.5m high)	Fair faced concrete painted	Built-up roofing waterproofing + Insulation	Cement mortar plaster painted	Steel flush door with semi-gloss paint finish	Aluminium window with 6 mm thick blue tinted polished float glass with reflective out board
11	Firefighting pump house	Plant Room (Pump room)	Monolithic steel trowel finish with hardener	Cement mortar plaster painted	Fair faced concrete painted	Uninsulated metal roofing	Cement mortar plaster painted	Steel flush double door with semi-gloss paint finish	Aluminium louver with insulation
12	Hydrogen Building		Monolithic steel trowel finish with hardener	Cement mortar plaster painted	Fair faced concrete painted	Uninsulated metal roofing	Cement mortar plaster painted	Steel flush double door with semi-gloss paint finish	Aluminium louver with insulation
13	Guard House	Room	Ceramic tile	Cement mortar plaster painted	Acoustic tile panel with aluminium suspension system	Built-up roofing waterproofing	Cement mortar plaster painted	Steel flush door with semi-gloss	Aluminium window with 6 mm thick blue tinted polished float glass with

ADD1
-IIG
-009

			Interior Finishes			Exterior Finish			
No.	Name of Building	Name of Room/ Floor	Floor	Wall	Ceiling / Cover	Roof	Wall	Door	Window
		Toilet	Unglazed ceramic tile	Glazed ceramic tile	Moisture re- sistance ceiling with aluminium sustension sys- tem	+ Insulation		paint finish	reflective out board

ADD1
-IIG
-010

APPENDIX-2 BUILDING STRUCTURE LIST

Building	Story	Structure
1. Steam Turbine Building	Multi	Steel rigid frame and bracing and steel roofing (Higher Roof) and build up roofing (Deaerator Floor Roof)
2. Central Control Building	Multi	RC structure
3. Cooling Water Pump House	Single	Steel rigid frame with bracing
4. Water Treatment Plant- Control Building	Single	Steel rigid frame with bracing
5. Waste Water Treatment Plant Control Building	Single	Steel rigid frame with bracing
6. Desalination Plant Control Building	Single	Steel rigid frame with bracing
7. Chlorination Building	Single	R.C. rigid frame
8. Emergency Diesel Generator Building	Single	RC. Structure
9. Storage room, Workshop and Laboratory	Multi	RC. Structure
10. Storage Building for B3 Waste and Material	Single	RC structure

ADD1
-IIG
-011

ADD1
-IIG
-012

Building	Story	Structure
11. Firefighting pump house	Single	Steel rigid frame with bracing
12. Hydrogen Building	Single	Steel rigid frame with bracing
13. Guard House	Single	RC structure

APPENDIX-3 BUILDING FACILITY LIST

Building	Room	Sanitary Fixture	Air Conditioning	Ventilation	Power Supply	Communication	Lightning and Grounding
1. Steam Turbine Building	Turbine Hall			○	○	○	
	Electrical Equipment Room			○ (Air Intake)		○	○
	Auxiliary Equipment Floor			○ (Air Intake)		○	
	Cable Trench				○		
2. HRSG Area (Heat Recovery Steam Generator)	Outdoor						○
3. Stack Area	Outdoor						○
4. Central Control Building	Central Control Room		○	○	○	○	○
	Electronic Equipment Room		○	○	○	○	○
	Storage			○	○	○	
	Meeting Room		○	○	○	○	
	Plant Manager Room and Technical Staff Office		○	○	○	○	
	Toilet & Wudhu	○		○	○		
	Library room		○	○	○	○	
	Prayer room		○	○	○	○	
	Locker room	○		○	○		
	Pantry	○		○	○		
	Air Conditioning Equipment Room			○	○		
	Battery Room			○	○		
	LV SWGR Area			○	○		
	MV SWGR Area			○	○		
5. Cooling Water Pump House	Electrical Room		○	○	○	○	○
	Plant Room			○	○	○	
6. Water Treatment Plant	Plant Room			○	○	○	○
	Control Room		○	○	○	○	
	Electrical Room		○	○	○	○	
	Toilet	○		○	○		

Building	Room	Sanitary Fixture	Air Conditioning	Ventilation	Power Supply	Communication	Lightning and Grounding
7. Waste Water Treatment Plant	Plant Room			○	○	○	○
	Control Room		○	○	○	○	
	Electrical Room		○	○	○	○	
	Toilet	○		○	○		
8. Desalination Plant	Plant Room			○	○	○	○
	Control Room		○	○	○	○	
	Electrical Room		○	○	○	○	
	Toilet	○		○	○		
9. Chlorination Building	Plant Room			○	○	○	○
	Control Room		○	○	○	○	○
10. Emergency Diesel Generator House	Plant Room			○	○	○	○
11. Storage room, Workshop and Laboratory Building	Workshop			○	○	○	○
	Office		○	○	○	○	
	Toilet	○		○	○		
	Laboratory	○	○	○	○	○	
12. Storage Building for B3 Waste and Material		○		○	○	○	○
13. Firefighting Pump house				○	○	○	○
14. Guard House		○	○	○	○	○	○

APPENDIX-4

SERVICE WATER SUPPLY (not including Operating/ Fire Fighting Requirement) & DRAINAGE CONCEPT

No.	Facility	Water Supply	Drainage Connection			
			Storm water Drainage	Sanitary Drainage System	Oily Water Drainage	Waste Water Drainage
1.	Steam Turbine Building	Floor Wash	Roof and sump pit	N/A	Floor Drains	Specific Drains, if any
2.	Transformer bay	N/A	Rain	N/A	Floor Drains	N/A
3.	HRSG Area	N/A	Roof and general area	N/A	Specific Drains	Specific Drains
4.	Central Control Building	Toilet Kitchen Re-stroom	Roof, Air-con drains	Sanitary drains	N/A	Specific Drains
5.	Cooling Water Pump House	Floor Wash	Roof	N/A	Floor Drains	N/A
6.	Water Treatment Building	Floor Wash Toilet Laboratory	Roof, Air-con drains	Sanitary drains	N/A	Specific Drains
7.	Chlorination Building	Floor Wash	Roof	N/A	Floor Drains	Specific Drains, if any
8.	Emergency Diesel Generator Building	Floor Wash	Roof	N/A	Floor Drains	N/A
9.	Storage room, Workshop and Laboratory Building	Toilet Kitchen	Roof, Air-con drains	Sanitary drains	N/A	Specific Drains
10.	Roads & Drains	N/A	Roads and surrounding area	N/A	N/A	N/A
11.	Cable/Pipe Trenches	N/A	Drain point	N/A	N/A	N/A
12.	Firefighting Pump House	N/A	Roof	N/A	Floor Drains	N/A
13.	Hydrogen Building	N/A	Roof	N/A	Floor Drains	N/A
14.	Guard House	Toilet	Roof, Air-con drains	Sanitary drains	N/A	N/A